Re-implementing PatchTST and Applying It to Retail Demand Forecasting

Yanlei He, Lijuan Luo
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1 Introduction

PatchTST [1] revisits Transformer based long horizon time series forecasting through two ideas: patching and channel independence. The paper argues that earlier Transformer forecasters often underperformed a simple linear baseline, DLinear [2], and that a better tokenization scheme can recover strong performance. Our project had two goals. First, we aimed to reproduce the main supervised PatchTST results. Second, we tested whether the same model transfers to a practical business task, retail demand forecasting on Favorita grocery sales.

2 Chosen Result

We focused on two central results from PatchTST [1]. The first is Table 3, which compares supervised PatchTST/42 against DLinear and a vanilla Transformer on ETTh1, ETTm1, and Weather. The second is Table 7, which ablates patching and channel independence. These results were chosen because they directly support the paper's two main claims: PatchTST improves forecasting accuracy, and both architectural choices matter. Figure 1 summarizes the benchmark trend, and Table 1 compares our results against the paper.

3 Methodology

Implementation. We implemented supervised PatchTST in PyTorch from the paper. The model uses RevIN [3], patch length $P = 16$, stride $S = 8$, look back window $L = 336$, and a shared Transformer encoder. Channel independence is implemented by reshaping the input from (B, L, M) to $(B \cdot M, L)$ so each channel is processed independently with shared weights. We also implemented DLinear [2], a vanilla Transformer baseline, and the ablation variants `P+CI`, `CI Only`, `P Only`, and `Original`.

Datasets and metrics. We used ETTh1, ETTm1, and Weather for replication. For the retail extension, we selected the largest Quito store from the Favorita dataset [4] and treated its 33 product families as channels. We trained with MSE loss, Adam, OneCycleLR, and early stopping. Results are reported with test MSE and MAE in standardized space, following [1].

Implementation challenges. Cross checking against the official code revealed three details that materially affected accuracy. First, PatchTST/42 uses replication padding that produces 42 patches rather than 41. Second, the encoder uses BatchNorm rather than LayerNorm. Third, DLinear uses shared weights across channels by default. Fixing these details closed an early 5 to 15 percent performance gap in our implementation.

4 Results and Analysis

Benchmark replication. Our results are best described as a partial replication. On Weather, PatchTST closely matched the paper and was best at all four horizons. On ETTm1, PatchTST also beat DLinear and the vanilla Transformer at all four horizons, but its absolute MSE remained above the paper. On ETTh1, PatchTST only partially matched the paper: it beat DLinear at horizon 96, but DLinear was slightly better at 192, 336, and 720. Figure 1 shows the benchmark trend lines for all three datasets, and Table 1 gives the detailed paper versus ours comparison.

Seed sensitivity. To test whether the ETT mismatch was just random initialization noise, we ran a 3 seed sweep on ETTh1 and ETTm1. The standard deviation across seeds was only 0.001 to 0.004, far smaller than the gap to the paper. This suggests that the remaining mismatch is systematic rather than random. Our interpretation is that the gap comes from implementation sensitive training details and script level hyperparameter overrides in the official repository.

Ablation. The full model, `P+CI`, performed best in every ablation setting we tested. This supports the paper's central claim that both patching and channel independence contribute to the final forecasting gain. Figure 2

shows the Weather ablation at horizon 336.

Retail extension. On Favorita, PatchTST achieved the best MSE at 7 day, 14 day, and 30 day horizons. It reduced MSE by about 27 to 35 percent relative to DLinear and outperformed the vanilla Transformer by a much larger margin. Figure 3 shows the retail horizon comparison. Figure 4 shows a 30 day forecast example for BEVERAGES. One interesting observation is that the retail horizon difficulty is not strictly monotonic. The 30 day MSE is slightly lower than the 14 day MSE. We interpret this as a dataset specific effect driven by seasonality and window level volatility rather than a replication error.

5 Reflections

This project showed that reproduction quality depends heavily on hidden implementation details. Even when the model design is correct, padding, normalization, and baseline defaults can materially change the results. A second lesson is that matching the exact numbers in a paper is different from reproducing its main conclusions. In our case, Weather matched closely, ETTm1 reproduced the correct ranking, and ETTh1 only partially matched. Still, the core value of PatchTST's architecture remained visible.

The extension to Favorita suggests that PatchTST is useful beyond benchmark datasets. A natural next step would be to add exogenous retail variables such as promotions, holidays, and oil prices, since the current model is purely autoregressive.

References

References

[1] Yuqi Nie, Nam H. Nguyen, Phanwadee Sinthong, and Jayant Kalagnanam. A Time Series is Worth 64 Words: Long-term Forecasting with Transformers. In *International Conference on Learning Representations*, 2023.

[2] Ailing Zeng, Muxi Chen, Lei Zhang, and Qiang Xu. Are Transformers Effective for Time Series Forecasting? In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2023.

[3] Taesung Kim, Jinhee Kim, Yunwon Tae, Chulhee Park, Jang-Ho Choi, and Jaegul Choo. Reversible Instance Normalization for Accurate Time-Series Forecasting against Distribution Shift. In *International Conference on Learning Representations*, 2022.

[4] Corporación Favorita. Store Sales: Time Series Forecasting. Kaggle competition. https://www.kaggle.com/competitions/store-sales-time-series-forecasting. Accessed May 2026.

Table

Dataset / T	Paper PatchTST	Ours PatchTST	Ours DLinear	Ours Transformer
ETTh1 / 96	0.375	0.442	0.446	0.622
ETTh1 / 192	0.414	0.492	0.491	0.754
ETTh1 / 336	0.431	0.548	0.539	0.922
ETTh1 / 720	0.449	0.659	0.642	1.127
ETTm1 / 96	0.290	0.346	0.363	0.591
ETTm1 / 192	0.332	0.396	0.409	0.637
ETTm1 / 336	0.366	0.438	0.454	0.660
ETTm1 / 720	0.420	0.500	0.512	0.757
Weather / 96	0.152	0.150	0.175	0.173
Weather / 192	0.197	0.196	0.218	0.217
Weather / 336	0.249	0.254	0.264	0.273
Weather / 720	0.320	0.319	0.328	0.334

Table 1: **Baseline replication results.** Comparison between the original PatchTST paper and our re implementation on ETTh1, ETTm1, and Weather. Lower MSE is better.

Figures

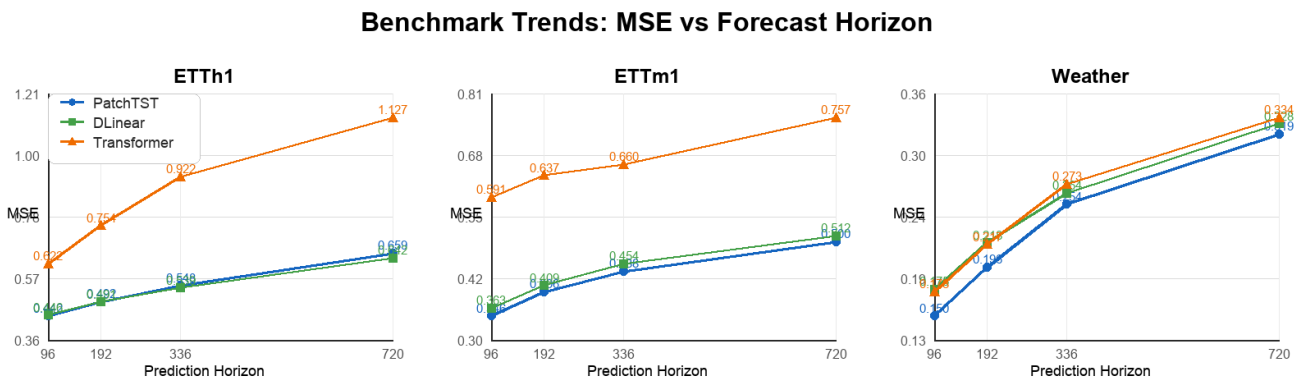

Figure 1: **Benchmark trend lines.** PatchTST is compared with DLinear and a vanilla Transformer on ETTh1, ETTm1, and Weather. The figure shows that PatchTST remains much stronger than the vanilla Transformer and reproduces the main ranking trends reported in the paper.

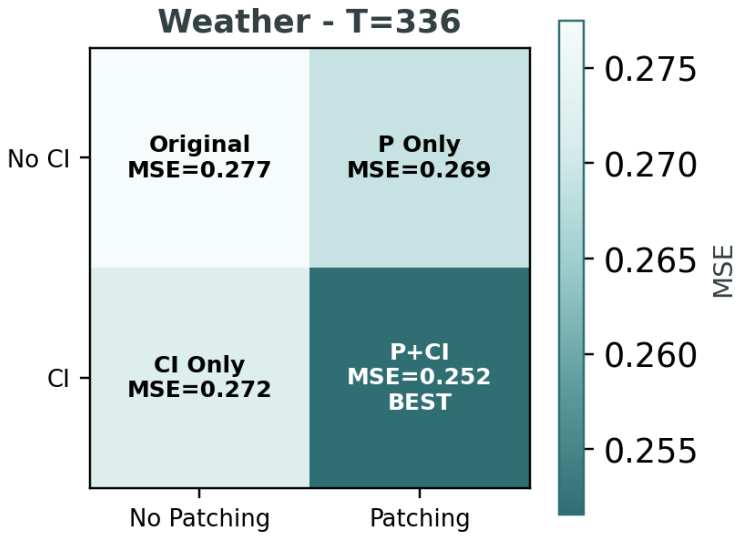

Figure 2: **Ablation on Weather at horizon 336.** The full model with patching and channel independence achieves the best MSE, which supports the paper's main architectural claim.

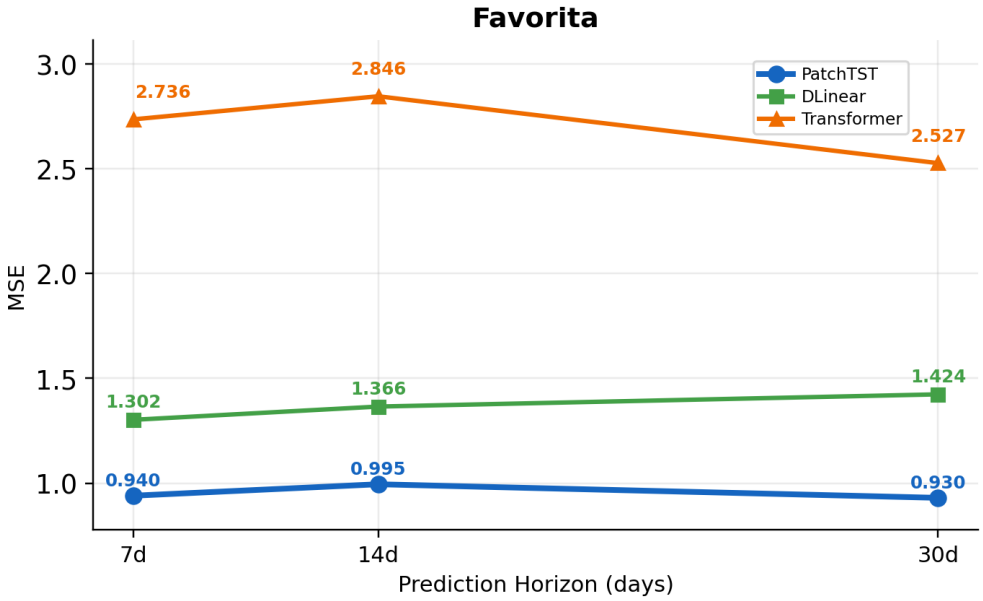

Figure 3: **Marketing extension on Favorita.** PatchTST achieves the lowest MSE at 7 day, 14 day, and 30 day retail forecasting horizons.

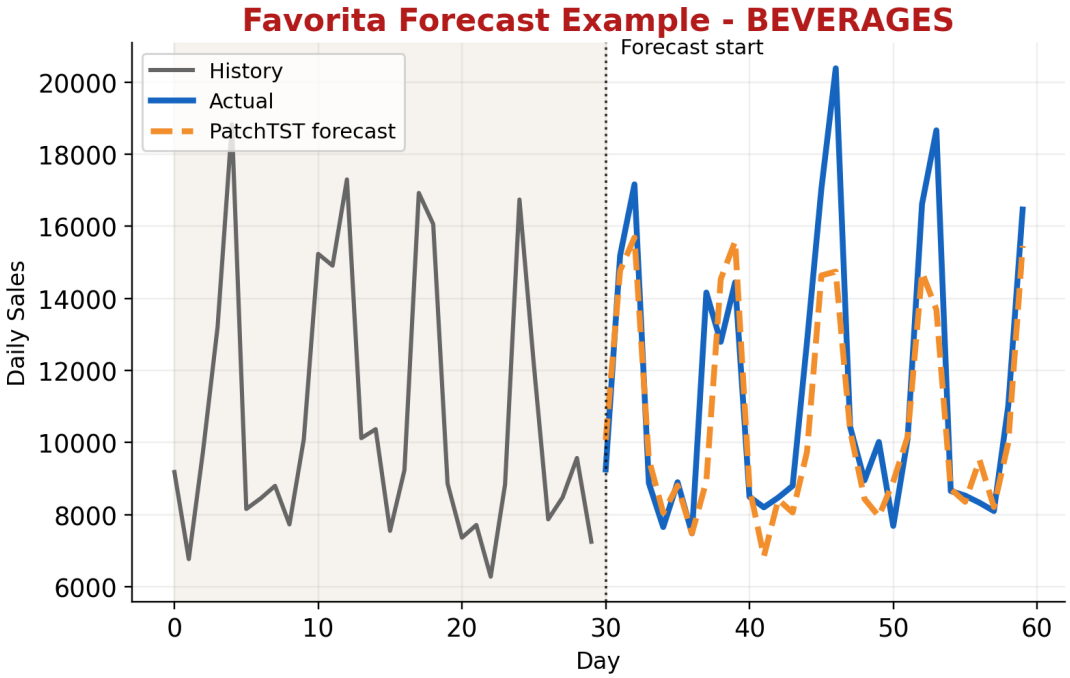

Figure 4: **Thirty day forecast example on BEVERAGES.** The gray line is the history, the blue line is the actual future, and the orange dashed line is the PatchTST forecast.